

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 02
Duration: 20 Minutes

Semester : Spring 25
Full Marks: 15

CSE420: Compiler Design

Name: Solution	ID:	Section:07
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1. Find out the terminals and nonterminals of the given Context Free Grammar: (10)
 - a. $S \rightarrow A - B C \mid BC$
 - b. $A \rightarrow a + [A] \mid \epsilon$
 - c. $B \rightarrow b B \mid b$
 - d. $C \rightarrow \{ c C \} \mid (d^*$
2. What are the possible actions of Shift-Reduce Parsing? (5)

Question 01:

TERMINALS: - , a , + , [,) , b , { , c , } , (, d , *

NON- TERMINALS: A , B , C , S(Start State)

Question 02:

While the primary operations are shift and reduce, there are actually four possible actions a shift-reduce parser can make: (1) shift, (2) reduce, (3) accept, and (4) error.

1. Shift. Shift the next input symbol onto the top of the stack.
2. Reduce. The right end of the string to be reduced must be at the top of the stack. Locate the left end of the string within the stack and decide with what nonterminal to replace the string.
3. Accept. Announce successful completion of parsing.
4. Error. Discover a syntax error and call an error recovery routine.